

Algebra 1
Unit 11
Lesson 6 Practice

Name Answer Key
Date _____
Period _____

Use the situation below to answer questions 1-3.

A new MacBook computer costs \$950, but depreciates 10.8% each year as new technology comes out.

1. Is this situation exponential growth or decay?
Exponential decay

2. Two students, Quinn and Jyquis were debating how to represent a function to model the value, V , after x years. Both of their functions are below. Explain who you agree with and why.

Quinn	Jyquis
$V(x) = 950(1.108)^x$	$V(x) = 1275(0.892)^x$

I agree with neither of them. Quinn has exponential growth in her equation, and Jyquis has the incorrect initial amount.

3. Write the correct function that models the context using both forms.

$V(x) = a(b)^x$	$V(x) = a(1 \pm r)^x$
$V(x) = 950(0.892)^x$	$V(x) = 950(1 - 0.108)^x$

Use the situation below to answer questions 4-6.

A new school opened with 134 students in 2021 and plans to increase by 26% per year until they reach full capacity.

4. Is this situation exponential growth or decay?
Exponential growth

5. Write an equation that models the population of the school, P , after x years since the opening of the school in 2021.

$$P(x) = 134(1.26)^x \text{ or } P(x) = 134(1 + 0.26)^x$$

6. Write the correct function that models the context using both forms.

$P(x) = a(b)^x$	$P(x) = a(1 \pm r)^x$
$P(x) = 134(1.26)^x$	$P(x) = 134(1 + 0.26)^x$

7. A three-bedroom house in the suburbs of Miami sold for \$420,625. If the housing prices are expected to increase 3.75% annually, write a function that models the cost of a house, C , x years after it was sold.

$$C(x) = 420,625(1 + 0.0375)^x \text{ or } C(x) = 420,625(1.0375)^x$$

8. A small local company reported a profit of \$89,000 last year but is struggling due to the pandemic. They are reporting a decrease of 12% each year. Write a function that represents the profit, P , after t years.

$$P(t) = 89,000(1 - 0.12)^t \text{ or } P(t) = 89,000(0.88)^t$$

9. As of 2020, the world population is 9.23 billion people and is expected to grow at a rate of 0.62% per year. Write an equation to model the population growth, where $P(y)$ is the population in billions of people and y is the time in years.

$$P(y) = 9.23(1 + 0.0062)^y \text{ or } P(y) = 9.23(1.0062)^y$$

10. A culture of 15 bacteria grows at a rate of 250% each hour. Tyler wrote the equation $B(t) = 15(2.5)^t$ to represent the population of the bacteria after t hours. Describe and correct the error that Tyler made.

Tyler forgot to add the 1 or 100% to the growth rate.

$$B(t) = 15(1 + 2.5)^t \text{ or } B(t) = 15(3.5)^t$$